3 A training & fine-tuning pipeline

Months 3–4

WHAT YOU WILL DO

A standardized training pipeline on open models: supervised fine-tuning, LoRA/QLoRA, and DPO alignment as the use case requires — with experiment tracking, checkpoints, and GPU cost discipline. This is also where efficiency becomes architectural: what attention costs, why the KV cache rules serving, and how the labs shrank both.

WHAT YOU WILL HAVE BUILT

An end-to-end, reproducible fine-tuning pipeline, with at least one model adapted to a pilot domain.

MATERIALS

BOOK — written

Ch. 13 Making attention cheap (FlashAttention · MQA · GQA · MLA · sliding windows)

INTERACTIVE

The Attention Zoo — five mechanisms, one decoding loop

IN PROGRESS

Ch. 16 Sequence packing · Ch. 22 SFT done cheaply

CH. 13

Making attention cheap

FlashAttention, MQA, GQA, MLA, sliding windows — and how to choose

THE CHAPTER IN FIVE IDEAS

- Attention presents three different bills: quadratic FLOPs (bind past $L^* \approx 6d \approx$ 25K tokens at 7B), the L×L training-memory artifact, and decode bandwidth — reading the whole KV cache per generated token at ~1 FLOP per byte.
- FlashAttention erases the memory artifact exactly — tiling through SRAM with an online softmax — and is simply always on.
- The cache shrinks by consolidating KV heads: MQA (÷32, quality tax), GQA-8 (the industry default; reachable from MHA by a 5% uptrain), or DeepSeek's MLA — cache a 576-number latent, attend on it directly: 57×, quality intact.
- Sliding windows cap compute and cache but trade away direct long-range sight — so Mistral and Gemma interleave local and global layers.
- Decision: Flash always → GQA-8 by default → MLA when serving memory is the product → windows when context is long and local. Then evaluate long-range recall — the currency every technique quietly spends.

550 GB vs 13.5 GB

the KV cache versus the weights for an MHA 7B at 32K context, batch 32 — why this chapter exists.

GO DEEPER

Read: chapters/ch13_attention.html
Lab (Colab): labs/ch13_attention.ipynb
Play: the Attention Zoo — five mechanisms, one loop